BTEC Level 3 Applied Science · Unit 1 Biology · Worksheet

Microscopy

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: magnification, resolution, conversions and instrument choice.

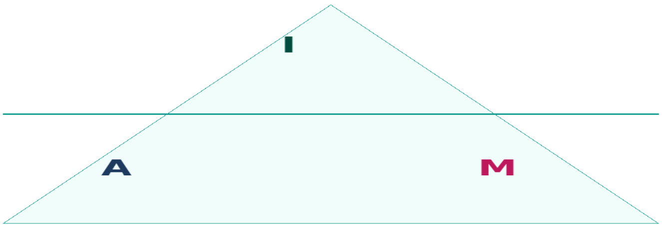
A Retrieval

1. Write the magnification formula and define resolution.

2. Convert 2.5 mm to µm and to nm.

Working

Magnification triangle



I = image size A = actual size M = I / A

Convert to the same unit first. 1 mm = 1000 µm = 1 000 000 nm

3. How do you calculate total magnification of a light microscope?

C Calculations — show every conversion

4. An image is 18 mm wide and magnification is ×600. Calculate actual size in µm.

Working

5. A 7.5 µm cell is drawn 3.0 cm wide. Calculate magnification.

Working

6. A 10 µm scale bar measures 40 mm on a photograph. Calculate magnification.

Working

D Exam-style practice

7. Calculate the actual width of a cell if the image is 35 mm wide at ×500. Give the answer in µm. [2]

Working

8. Compare the resolution of a light microscope and a TEM. [4]

9. Give two reasons for choosing a light microscope rather than an electron microscope. [2]

E Challenge

10. A student increases magnification from ×100 to ×1000 but cannot separate two close granules. Explain, using resolution.